

Inventory Management Console Application in C#

Objective:

Develop a console-based inventory management application in C#. This application will allow users to manage a list of products, including adding, editing, viewing, and deleting items. This task will enhance your skills in object-oriented programming, data handling, and console interface design.

Requirements:

Product Information:

- Allow users to enter product details, including product ID, name, price, and quantity in stock.
- Each product should be represented as an object of a Product class.

Inventory List:

- Maintain an in-memory list of products.
- Provide functionalities to add new products, edit existing ones, delete products, and view the list of all products.

Search Functionality:

- Implement a search feature to find products by name or ID.
- Display relevant product details or a message if no matching product is found.

User Interface:

- Design a console-based user interface to interact with the inventory system.
- Utilize console input/output methods for displaying and manipulating product information.
- Implement a simple menu system for accessing different functionalities.

Error Handling:

- Include proper error handling to prevent crashes from invalid inputs.
 - Use try-catch blocks to manage exceptions and display user-friendly error messages.

Data Persistence:

- Since this is a basic version, data persistence beyond the application's runtime is not required.
- Focus on in-memory data management for this version.

User Experience:

- Ensure smooth and intuitive interactions within the console environment.
- Consider adding features like sorting the inventory list or providing quick access commands.

Documentation:

- Provide clear instructions on how to run and interact with the application.
- Document the functionalities and any special features or commands.

Code Quality:

- Follow good coding practices with clear naming conventions and code structure.
- Write clean, readable, and well-commented code to ensure maintainability.

